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HZSM5 and HUSY deactivation during the catalytic pyrolysis of polyethylene

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Abstract

The behaviour of HZSM5 and HUSY zeolites during the decomposition of low density (LDPE) and high density polyethylene (HDPE) has been studied by TGA. Zeolites have been thoroughly characterized in order to elucidate the structures responsible for the observed behaviour. The reduction in the decomposition temperature is initially very high with HUSY for both polyethylenes while for HZSM5 is much greater with the branched LDPE than with linear HDPE. To explain the relative activity of these zeolites and the differences found for HZSM5 with both polyethylenes the polymer structure, the zeolite acidity and its textural properties have to be considered. The progressive deactivation behaviour of the catalysts has also been studied. HUSY shows a fast deactivation while small pore size in HZSM5 prevents coke deposition. Completely deactivated zeolites still show a considerable number of apparently unaltered acid sites.

Keywords: Catalytic pyrolysis; Catalyst activity; Deactivation; Coke deposition; HZSM5; HUSY; HDPE; LDPE

1. Introduction

Catalytic pyrolysis of waste polymers is being recognised as a new recycling approach. The reduction in the decomposition temperature as well as the possibility to obtain chemicals or fuels could significantly reduce the cost of the recycling process. There are a considerable number of works dealing with mechanistic considerations of the thermal degradation of polymers, and with catalytic degradation of low molecular weight compounds on acid catalysts [1–4]. But only a few of them deal with mechanisms of catalytic degradation of polymers.

For the catalytic decomposition of heavy hydrocarbons, in spite of different mechanisms being proposed, it is agreed that the reactant molecules are adsorbed on the acid sites of the catalysts where they are protonated to obtain carbenium ions, which promote the cracking of the molecule. In sequent steps hydrogen transfer reactions, isomerization or β-scission reactions, etc., take place to produce a distribution of low molecular weight compounds. For large molecules as is the case of polymers, it has been suggested [5,6] that decomposition reactions are initiated on the active sites located at the external surface of the zeolite. Lower molecular weight products obtained can diffuse through the molten polymer as products or react in the pores and micropores where smaller molecules are obtained.

Whether in the external surface or in the pores, it is generally agreed that decomposition reactions are initiated on the acid sites of the catalysts. Consequently acid sites type, acid strength and distribution are clue characteristics in the behaviour of catalysts during the catalytic decomposition of polymers. In zeolites Brönsted and Lewis acid sites can be found related to intra-framework and extra-framework aluminum. There are a considerable number of works dealing on the strength and location of these acid sites but some discrepancies can be found in the literature.

For the HZSM5 zeolite different authors [7–10] obtained that the strongest Brönsted acid sites are located in the inner surface of pores while external acid sites correspond mainly to silanol groups of medium to high acidity. According to

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this author external groups are involved in reactions with hydrocarbons only once internal groups are saturated. Drago et al. [7] found no measurable Brönsted acid sites in the external surface of HZSM5. Nevertheless Zheng [11] suggested that a fraction of Lewis acid sites are located at the external surface of HZSM5 and that these sites were probably stronger acidic than Brönsted acid sites.

For the HUSY zeolite Guisnet and coworkers [12] found that the concentration of Brönsted acid sites was double that of Lewis acid sites but the acid strength of the last ones was higher than that of Brönsted sites. Nevertheless, other researchers concluded [13–16] that in alcane cracking only the strongest Brönsted acid sites are active. Kung and coworkers [17] found that strong Lewis acid sites in HUSY do not affect the cracking activity of the catalyst. Nevertheless, Abbott [18] proposed that Lewis acid sites promote the chain reaction process of cracked products on the surface, which leads to coke formation.

The deactivation of the catalysts by coke deposition is one of the main disadvantages of the catalytic pyrolysis of polymers. Deactivation of catalysts is a complex phenomenon and can be due to different causes: poisoning of active centres, fouling, thermal degradation and corrosion or leaching by the reaction mixture [19,20]. For zeolites employed in catalytic pyrolysis of polymers, the deactivation process is mainly due to active site poisoning and pore blockage by coke deposition [21]. The coke formation on different catalysts was studied by Guisnet and Magnoux [22] at low and high temperatures. A strong dependence of coke composition with temperature was observed. At low temperatures, these deposits are generally non-polyaromatic and their composition depends very much on the reactants. Nevertheless, at high temperatures, the coke composition is independent of the reactant type and the coke composition is polyaromatic. The rate of coke formation depends mainly on the zeolite structure. Coke molecules tend to be preferentially produced on the strongest acid sites.

In this work the reduction of the decomposition temperature when HZSM5 and HUSY zeolites are employed during the decomposition of low density and high density polyethylene (LDPE and HDPE, respectively) have been studied. The behaviour of the catalysts during the progressive deactivation has also been studied. Catalysts have been thoroughly characterized prior to and after deactivation in an attempt to obtain information on the type of structures responsible for the activity of the zeolites.

2. Equipment and experimental procedure

Two commercial polyethylene samples were employed: low density polyethylene LDPE 780R supplied by DOW (LDPE) and high density polyethylene HDPE HD3560UR supplied by BP (HDPE). Both polymers were supplied in powder form with a maximum particle size of 500 μm. The melt flow index (MFI) was determined for LDPE and HDPE, 20 and 6 g at 10 min^{-1}, respectively. MFI is determined as the quantity of polymer extruded through a 1 mm die at 190 °C, under a weight of 2.16 kg and for 10 min. MFI is a measure of viscosity at zero shear rate and so, for polymers of similar structure, it is an indirect measurement of molecular weight. Density is determined at ambient temperature and it is well known that it is mainly related to crystallinity and this one to branching degree. LDPE is a branched polymer (0.923 g cm^{-3}) while HDPE is a more crystalline non-branched polymer (0.935 g cm^{-3}).

Commercial HZSM5 and HUSY zeolites from GRACE-Davison were employed as catalysts. These zeolites were chosen due to the differences in their properties. HUSY possesses greater aluminum content, much higher surface area and micropore volume than HZSM5. In fact HZSM5 is a zeolite with two types of channels: zigzag channels (5.3 Å × 5.6 Å) and linear channels (5.1 Å × 5.5 Å), while HUSY present supercages (7.4 Å).

Zeolites were characterized according to their surface area, porosity, and relative strength and distribution of their acid sites. Nitrogen adsorption–desorption isotherms at 77 K were performed using AUTOSORB-6 equipment. Samples were preheated at 250 °C for 5 h under vacuum. The Brunauer–Emmett–Teller (BET) procedure was applied to obtain surface area. The external surface area and the micropore volume were calculated by application of the *t*-plot method to a selected zone of the isotherm. The relative strength and distribution of acid sites were studied by temperature programmed desorption (TPD) in a Netzsch TG 209 thermobalance. The samples are heated under a flowing nitrogen atmosphere to 500 °C at 10 °C min^{-1} and then cooled to 100 °C and treated with a mixture of 5% ammonia in nitrogen flow for 30 min. The desorption of physisorbed ammonia takes place in nitrogen atmosphere for an hour at 100 °C. Then the temperature is increased from 100 to 900 °C at 10 °C min^{-1} to remove the chemically bonded ammonia. Acidity of the zeolites is expressed as mmol of ammonia removed per gram of zeolite.

The study of catalyst behaviour and cycle life was performed in a Netzsch TG 209 thermobalance. A calibration procedure was performed each day using internal standards (aluminum, nickel and perkalloy). The calibration with the three metals or alloys was recorded with a magnet surrounding the oven and sample. The standards lose their magnetic properties at a specific temperature (Curie-point transition at 163, 354 and 597 °C, respectively). In this way, the reliability of measured temperatures was assured. Samples were swept with nitrogen with a flow rate of 30 ml min^{-1}. Experiments were carried out from 30 to 550 °C at 10 °C min^{-1}. In all experiments, the catalyst to polymer ratio was 1/10. Lin et al. [23] studied different procedures of sample preparation and concluded that the dry mixing of the powder polymer with the catalyst led to a good reproducibility of the results. Accordingly thorough mixing of the powdered polymer and the catalyst was carried out prior to all experiments and good reproducibility was also obtained.

In order to check the remaining relative activity of the partially coked catalyst obtained in the experiments described above, fresh polymer was added maintaining the catalyst to polymer ratio in 1/10. This procedure was repeated until the complete deactivation of the catalyst. A similar procedure was carried out by Lin et al. [23] to quantify deactivation behaviour of HUSY with HDPE.

The nomenclature employed in the deactivation experiments is the following: the first letter in each run corresponds to the polymer, "L" and "H" for LDPE and HDPE, respectively; the second letter "Z" and "U" corresponds to the zeolite employed, HZSM5 and HUSY, respectively. The last digit corresponds to the number of times virgin polymer has been added to the catalyst.

Partially coked zeolites obtained after the first deactivation run (LZ1, HZ1, LU1 and HU1) and after the last run (completely coked) (LZ13, HZ6, LU5 and HU4) were also characterized by the techniques described above. The quantity of coke deposited on these zeolites was measured in a Perkin Elmer TGA1 thermobalance. These experiments were carried out in oxygen atmosphere at 30 ml min^{-1} and temperatures were varied from 30 to 900 °C at 10 °C min^{-1}. The coke deposited on the zeolite is expressed as weight percentage respect to the zeolite.

3. Results and discussion

3.1. Relative activity of the zeolites and deactivation behaviour by TGA

Fig. 1 shows the TG and DTG curves of LDPE and HDPE and those corresponding to the mixtures of these polyethylenes with HUSY and HZSM5 during the first deactivation run. Pure polyethylenes decompose at very similar temperatures and, as expected, LDPE decomposes at slightly lower temperatures than HDPE. This behaviour has been previously reported by different authors [24–28], and it seems to be related to the different degree of branching of both polyethylenes. The tertiary carbons present in the branched LDPE favour free radical generation resulting in a lower thermal stability.

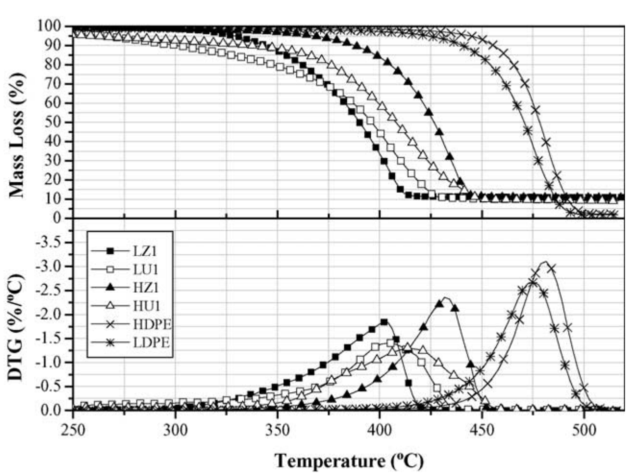

Fig. 1. TG and DTG curves of pure branched LDPE and linear HDPE and the corresponding mixtures with HUSY and HZSM5 during the first run.

Table 1
Temperature of maximum decomposition rate and ΔT_{max} of branched LDPE and linear HDPE and mixtures of both polymers with HUSY

Experiment	T_{max} (°C)	ΔT_{max} (°C)	Experiment	T_{max} (°C)	ΔT_{max} (°C)
LDPE	475.0	–	HDPE	481.5	–
LU1	405.4	69.6	HU1	413.7	67.8
LU2	457.0	18.0	HU2	469.7	11.8
LU3	459.2	15.8	HU3	475.2	6.3
LU4	467.4	7.6	HU4	477.5	4.0
LU5	467.3	7.7			

HUSY mixtures are represented by white dots in Fig. 1. The observed residues correspond to the catalysts and to the small quantity of coke deposited on them. As will be shown later, coke content was always lower than 20.2% with respect to the zeolite, which represents less than 2% with respect to the polymer. The initial decomposition temperature for HUSY mixtures is as low as 250 °C for both polyethylenes, but decomposition takes place at a low rate (a low slope is observed). A look at the DTG curves shows broad curves where different processes might be involved and the curve for HDPE is broader than for LDPE, showing a shoulder at the higher temperatures. Table 1 shows the temperature of the maximum decomposition rate for all experiments performed with HUSY. The difference between the temperature of maximum decomposition rate of the pure polymer and that of the mixture of the corresponding polymer with the catalyst (ΔT_{max}) is also presented in Table 1. ΔT_{max} has been considered as a measure of the relative activity. When HUSY is employed the ΔT_{max} is very similar for both polyethylenes (69.6 and 67.8 °C for LU1 and HU1, respectively). Apparently the ΔT_{max} of HUSY is not greatly influenced by the type of polymer employed.

Table 2
Temperature of maximum decomposition rate and ΔT_{max} of branched LDPE and linear HDPE and mixtures of both polymers with HZSM5

Experiment	T_{max} (°C)	ΔT_{max} (°C)	Experiment	T_{max} (°C)	ΔT_{max} (°C)
LDPE	475.0	–	HDPE	481.5	–
LZ1	402.2	72.8	HZ1	431.7	49.8
LZ2	414.8	60.2	HZ2	455.9	25.6
LZ3	424.1	50.9	HZ3	464.7	16.8
LZ4	426.1	48.9	HZ4	470.7	10.8
LZ5	428.1	46.9	HZ5	473.0	8.5
LZ6	430.0	45.0	HZ6	474.0	7.5
LZ7	436.0	39.0			
LZ8	438.1	36.9			
LZ9	445.9	29.1			
LZ10	444.0	31.0			
LZ11	446.0	29.0			
LZ12	444.2	30.8			
LZ13	446.1	28.9			

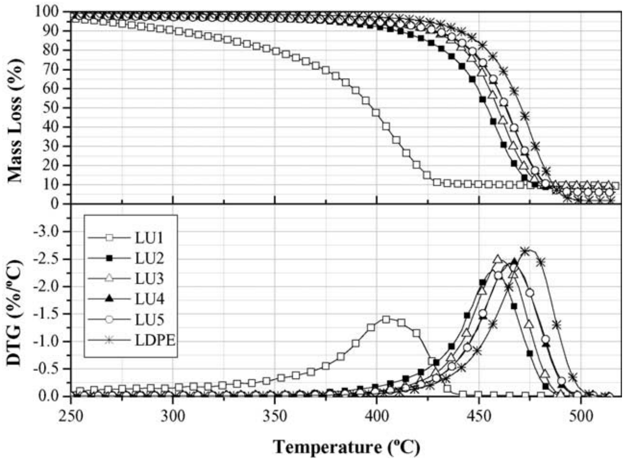

Fig. 2. TG and DTG curves of branched LDPE and mixtures of branched LDPE and HUSY during consecutive experiments adding fresh polymer to the coked catalyst.

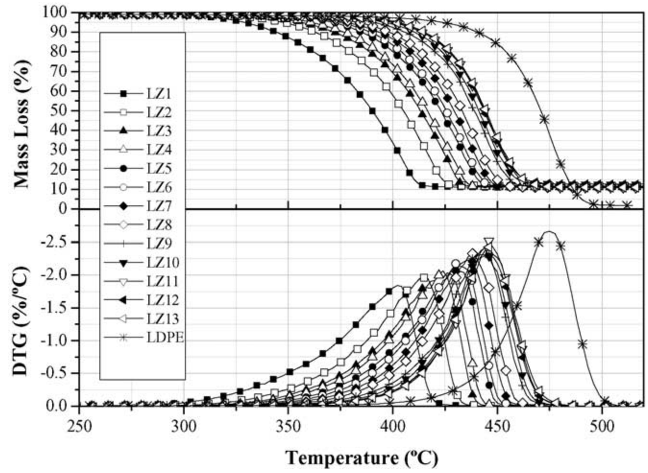

Fig. 4. TG and DTG curves of branched LDPE and of mixtures of branched LDPE and HZSM5 during consecutive experiments adding fresh polymer to the coked catalyst.

Nevertheless, the relative activity of HZSM5 is very different with both polyethylenes (solid dots in Fig. 1). The reduction in the decomposition temperature is 72.8 °C for LDPE and 49.8 °C for HDPE (Table 2 and Fig. 1). During the first run HZSM5 is much more active with LDPE than with HDPE. The shape of the DTG curve of HZ1 is quite similar to that obtained for the thermal decomposition of HDPE.

As commented on previously, successive runs were carried out where fresh polymer was added to the coked catalyst. In this way the catalyst cycle life was studied. Figs. 2 and 3 show the TG and DTG curves for HUSY with LDPE and HDPE, respectively. Temperatures of maximum decomposition rate and ΔT_{max} are shown in Table 1. As observed before, both polyethylenes again behave in a very similar way to HUSY. After the first run the loss in the zeolite relative activity is very high and an important change in the DTG curve can be observed, since DTG curves for the second run (LU2 and HU2) are more similar to those for the pure polyethylenes than to those for the first run (LU1 and HU1). After four runs in the case of HDPE and five runs for LDPE the zeolite is nearly completely deactivated and the decomposition takes place as if there were no zeolite present.

Figs. 4 and 5 show the TG and DTG curves obtained for LDPE and HDPE with HZSM5, while Table 2 presents the resultant ΔT_{max}. Deactivation of HZSM5 is by far much slower than that of HUSY. There are again clear differences between the behaviour of LDPE and HDPE. HZSM5 loses the relative activity very slowly with LDPE. After 13 runs there is still a considerable remaining catalytic activity that seems to be stabilized. With HDPE the initial relative activity is lower than shown above, and the catalyst loses the relative activity after six runs. In both cases changes in the shape of DTG curves are progressive, tending forward to the thermal decomposition of polyethylene.

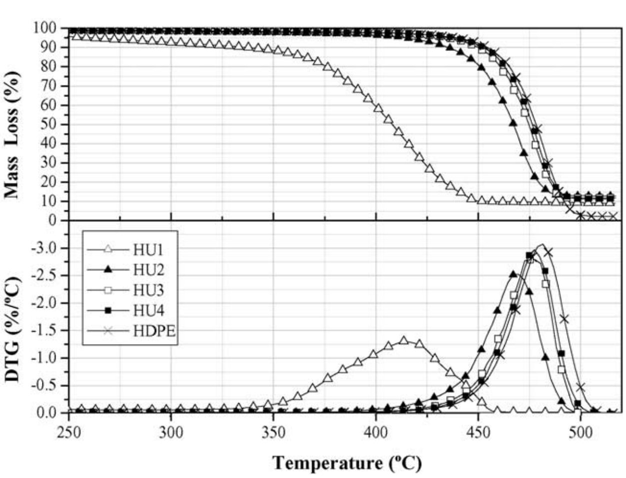

Fig. 3. TG and DTG curves of linear HDPE and mixtures of linear HDPE and HUSY during consecutive experiments adding fresh polymer to the coked catalyst.

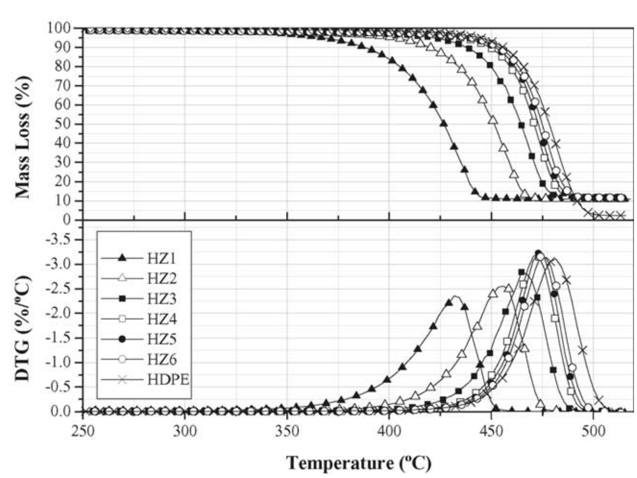

Fig. 5. TG and DTG curves of linear HDPE and of mixtures of linear HDPE and HZSM5 during consecutive experiments adding fresh polymer to the coked catalyst.

Table 3
Coke content and acidity for the pure and coked zeolites

Zeolite	Coke content respect to the zeolite (wt.%)	T_{max} (°C) (LT)	Acidity (mmol NH_3 g^{-1})	T_{max} (°C) (HT)	Acidity (mmol NH_3 g^{-1})
HZSM5	–	166	1.15	416	0.88
HZ1	3.6	166	1.16	416	0.76
HZ6	5.0	166	1.11	416	0.58
LZ1	5.4	166	1.12	416	0.66
LZ13	8.2	166	0.79	416	0.31
HUSY	–	154	2.12	–	–
HU1	14.0	154	1.67	–	–
HU4	18.0	154	1.53	–	–
LU1	16.0	154	1.55	–	–
LU5	20.2	154	1.34	–	–

3.2. Initial relative activity of both zeolites

In order to clarify the reasons for the behaviour commented above, the coked zeolites were characterized and compared to the pure zeolites. Table 3 shows the results of the coke content and the acidity for the pure and coked zeolites after the first deactivation run (HZ1, LZ1, HU1 and LU1) and after the last deactivation run (HZ6, LZ13, HU4 and LU5). Table 4 shows the textural properties of these zeolites.

HZSM5 presents two types of acid sites (Table 3), the low temperature sites (LT) with the maximum desorption of ammonia at 166 °C (1.15 mmol $NH_3\ g^{-1}$) and the high temperature sites (HT) at 416 °C (0.88 mmol $NH_3\ g^{-1}$). This behaviour of HZSM5 has been previously described by other authors [9,29] who attributed LT peaks to Brönsted and Lewis weak acid sites. As commented previously, the importance of weak acid sites on catalysis has sometimes been neglected. HT peaks have been associated to strong acid sites, by far more active. On the other hand, the acidity of HUSY has been described to be quite complex, with a wide distribution of the strength of its acid sites [30]. TPD of HUSY shows a very broad peak centred at 154 °C with an ammonia desorption of 2.12 mmol g^{-1}, which should correspond to a high number of relatively weak acid sites.

In general it is accepted for a zeolite that the higher the acidity, the higher the activity [11]. Obviously other considerations have to be borne in mind when zeolites of different families are compared. Serrano et al. [31] found a good correlation between the activity of a zeolite and its external surface area, and other authors [32,5] state that large macromolecules have to react on the external surface of the zeolite first. In this case since the acidity of HUSY is not very high as compared to HZSM5, the effect of textural properties was evaluated. HUSY presents a large surface area and pore volume, where most of the surface area is located at the pores, as shown in Table 4. Despite the not very high acidity, it has to be considered that the large pores of HUSY allow the molten polymer or the heavy initial products of the primary reactions to diffuse to the acid sites easily and so, the reactions are also initiated in the pore surface and not only in the external surface. In this case the high initial relative activity could be due to the easy penetration of the polymer to the acid sites.

3.3. Differences in the reactivity of LDPE and HDPE with HZSM5

The high acidity of HZSM5 may imply a high activity but big differences in the relative activity with LDPE and HDPE were found. To explain the behaviour of both polyethylenes in this case, the polymer and the zeolite structure have to be borne in mind.

Aguado et al. [33] stated that in the cracking of polyethylene, the effect of the external surface of the crystal of HZSM5 employed was negligible. In this polymer cracking takes place by penetration of molecules, at least to a certain chain length, into the zeolite pores and subsequent reaction over the internal acid sites. More recently Zhou et al. [34] confirmed that LDPE molecules were able to diffuse into the narrow pores of modified ZSM5, while polypropylene molecules, which present a major effective cross-

Table 4
Textural properties of the pure and coked zeolites after the first deactivation run

Parameter	HZSM5	HZ1	LZ1	HUSY	HU1	LU1
BET surface area ($m^2\ g^{-1}$)	341	300	211	614	252	168
External surface area ($m^2\ g^{-1}$)[a]	37.6	37.6	20.2	28.1	59.4	33.1
Micropore volume ($cm^3\ g^{-1}$)[a]	0.16	0.13	0.10	0.29	0.10	0.07
Pore volume ($cm^3\ g^{-1}$)[b]	0.18	0.16	0.11	0.35	0.14	0.11

[a] Obtained by application of the *t*-plot method.
[b] Measured at $p/p_0 = 0.99496$.

section, were not. Marcilla et al. [35] suggested the idea that the polymeric branches or chain ends may access to active sites located in the surface of larger pores of the zeolite, thus increasing the activity. According to this, the catalytic effect will be higher on polymers that show a higher number of branches or chain ends, since the decomposition would have a high number of initiation points. Due to the large pore size of HUSY, LDPE and HDPE chains find no obstacle to reach the active sites located in pores and, as it has been shown, the behaviour of both polyethylenes in the presence of HUSY is very similar. Nevertheless, HZSM5 presents a small pore size that may assume a steric impediment to the penetration of the polymer chains. Branched LDPE has a high number of chain ends that could reach the HZSM5 pores, so degradation mechanism is promoted. But HDPE is a linear not branched polymer, and the penetration of the polymer to the HZSM5 active sites is not promoted.

3.4. Coke deposition and cycle life of zeolites

During the evolution of the catalytic reactions, coke molecules are formed on the active sites. Coke deposits block active sites, or the catalysts pores and intersections, thus preventing the access of the reactant molecules to these sites. The changes that take place in the zeolite properties are always greater in the catalytic pyrolysis of LDPE than in that of HDPE as it does in the coke content (Table 3). This is observable for both zeolites but especially when HZSM5 is employed. LDPE molecular structure could improve the formation of coke molecules but also the easier penetration of LDPE to active sites located in pores commented above could be responsible for the greater extension of the catalytic reactions, and so the greater coke deposition.

A major deactivation and shorter cycle life was shown in previous sections for HUSY than for HZSM5. Accordingly, in Tables 3 and 4 a major coke content and major changes in the coked HUSY properties can be observed. It is known that coke molecules tend to be produced on the strongest acid sites [36]. In Table 3 it can be observed for HZSM5 as coke deposition is improved on HT peaks since these suffer a higher reduction after the first run than LT peaks (LZ1 and HZ1 peaks as compared to HZSM5). On strong acid sites, which are more active, coke deposition is more important and so, the initial effect of deactivation is more pronounced than in less acid active sites. Despite the presence of HT peaks in HZSM5, there is a low coke deposition for HZSM5, as can be seen in Table 3. The low coke deposition of HZSM5 could be due to its small pore size. In the formation of coke molecules bimolecular reactions are involved between bulky molecules (condensation, hydrogen transfer, etc.). The formation of these molecules in HZSM5 pores presents higher steric hindrances than in HUSY [22,36]. The short cycle life for HUSY is due to a fast deactivation by coke deposition, while small pore size in HZSM5 prevents coke deposits being formed and so, this zeolite presents long life cycles, despite its high acidity.

Table 4 shows the drastic changes in HUSY pore volume after the first deactivation run (LU1 and HU1 compared to HUSY) to the point that the total pore volume after the first run is of the order of that HZSM5 zeolites (LZ1 and HZ1). Thus, a reduction in pore size as a consequence of the coke deposition on the pore walls can be assumed. As shown in Figs. 2 and 3 HUSY is nearly deactivated after the first run and LDPE and HDPE behaviour was reported to be very similar. Nevertheless for sequent cycles, when the pore size of HUSY has been reduced as a consequence of coke deposition, the relative activity of HUSY is greater for LDPE than for HDPE (Table 1). This would confirm that differences in reactivity of both polyethylenes with zeolites are due to branching degree and are only observable for zeolites of a reduced pore size.

Table 3 also shows the acidity and coke content for the zeolites after the last deactivation run (HZ6, LZ13, LU5 and HU4 residues). It is worthwhile to remark that Table 3 shows a considerable number of acid sites present when the zeolites have been completely deactivated. According to other authors [9,29] in the case of HZSM5 LT peaks are practically untouched, except for LZ13. This could indicate that the type of acid sites responsible for the remaining acidity do not participate in the cracking reactions, or simply that these sites are not accessible to the polymer. Small NH_3 molecules are able to reach and to react with them, but greater molecules or polymer chain ends are not able to reach them.

A remarkable fact is that despite LZ13 residue still presents considerable relative activity (Table 2 and Fig. 4) the active sites have been drastically reduced if compared to the HZ6 residue, which shows a completely exhausted relative activity. This is indicative of two facts; on one hand LDPE is reacting with active sites at which HDPE is not able to accede, which has been commented previously, and what is more important, the HZ6 residue, which is exhausted when reacting with HDPE, could still present a high activity if reacting with LDPE.

4. Conclusions

Not only the zeolite structure but also the polymer structure determines the activity of a zeolite in the catalytic cracking of polymers. The polymer chain ends are able to penetrate into the zeolite pores, reaching acid sites located there in and increasing the activity. Branched polymers, as is the case of LDPE, show a greater relative activity with zeolites of small pore size (HZSM5) than non-branched polymers, while there are no differences in the relative activity of branched and non-branched polymers if the zeolite pores are large enough for the main chain to penetrate, as has been shown for HUSY. By other hand the higher the pore size the higher the initial relative activity, as has been shown for HUSY, but the coke deposition is improved in zeolites of large pores, leading to a fast deactivation.

Once the zeolite has been completely deactivated with a polymer there are still a considerable number of acid sites untouched, indicating that the zeolite could still present high activity if employed with the appropriate polymer.

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